

1

PHY 615-2012
PROBLEM SET #1

Problem 1 (14.3.1)

Consider spinor $\begin{pmatrix} p_1 e^{i\phi_1} \\ p_2 e^{i\phi_2} \end{pmatrix}$

Normalization

$$\Rightarrow p_1^2 + p_2^2 = 1$$

Parametrize

$$\begin{aligned} p_1 &= \cos \frac{\theta}{2} \\ p_2 &= \sin \frac{\theta}{2} \end{aligned}$$

$$\chi = e^{i \frac{\phi_1 + \phi_2}{2}} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i \frac{\phi_2 - \phi_1}{2}} \\ \sin \frac{\theta}{2} e^{i \frac{\phi_2 - \phi_1}{2}} \end{pmatrix}$$

$$\Rightarrow n \text{ is along direction } \theta, \phi \equiv \theta, \frac{\phi_2 - \phi_1}{2}.$$

Problem 2 (14.3.2)

$$(1) \hat{n} \cdot \vec{\sigma} = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}$$

$$(\hat{n} \cdot \vec{\sigma}) |\hat{n} + \rangle = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi/2} \\ \sin \frac{\theta}{2} e^{i\phi/2} \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\phi/2} (\cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2}) \\ e^{i\phi/2} (\sin \theta \cos \frac{\theta}{2} - \cos \theta \sin \frac{\theta}{2}) \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\phi/2} \cos \frac{\theta}{2} \\ e^{i\phi/2} \sin \frac{\theta}{2} \end{pmatrix} = |\hat{n} + \rangle$$

$$(\hat{n} \cdot \vec{\sigma}) |\hat{n} - \rangle = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\phi/2} \\ \cos \frac{\theta}{2} e^{i\phi/2} \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\phi/2} (-\sin \frac{\theta}{2} \cos \theta + \sin \theta \cos \frac{\theta}{2}) \\ e^{i\phi/2} (-\sin \theta \sin \frac{\theta}{2} - \cos \theta \cos \frac{\theta}{2}) \end{pmatrix}$$

$$= - |\hat{n} - \rangle$$

$$(2) \langle n + | \sigma_x | n + \rangle$$

$$= \left(\cos \frac{\theta}{2} e^{i\phi/2}, \sin \frac{\theta}{2} e^{-i\phi/2} \right) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\phi/2} \\ \sin \frac{\theta}{2} e^{i\phi/2} \end{pmatrix}$$

$$= \left(\sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{i\phi} + \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{-i\phi} \right)$$

$$= 2 \cos \phi \sin \frac{\theta}{2} \cos \frac{\theta}{2} = \sin \theta \cos \phi = n_x$$

Thus $\langle n+ | S_x | n+ \rangle = \frac{\hbar}{2} n_x$

The other relations can be verified similarly

Problem 3 (14.3.3)

From (14.3.32)

$$\text{Tr } \sigma_i \sigma_j = -\text{Tr}(\sigma_j \sigma_i) = -\text{Tr}(\sigma_i \sigma_j)$$

(Since $\text{Tr } AB = \text{Tr } BA$)

$$\Rightarrow \text{Tr } \sigma_i \sigma_j = 0 \quad (i \neq j)$$

From (14.3.33)

$$\sigma_i \sigma_j = i \epsilon_{ijk} \sigma_k$$

Thus we have

$$i \epsilon_{ijk} \text{Tr } \sigma_k = 0$$

Since ϵ_{ijk} is totally antisymmetric this implies

$$\text{Tr } \sigma_k = 0 \quad \forall k.$$

eg for $i=1, j=2$

$$i \epsilon_{12k} \text{Tr } \sigma_k = i \text{Tr } \sigma_3 = 0$$

Problem 4 (14.3.4)

$$\begin{aligned}
 (1) \quad (\vec{A} \cdot \vec{\sigma})(\vec{B} \cdot \vec{\sigma}) &= A_i B_j \sigma_i \sigma_j \\
 \sigma_i \sigma_j &= \frac{1}{2} [\sigma_i, \sigma_j] + \frac{1}{2} \{\sigma_i, \sigma_j\} \\
 &= i \epsilon_{ijk} \sigma_k + \delta_{ij} \\
 \Rightarrow (\vec{A} \cdot \vec{\sigma})(\vec{B} \cdot \vec{\sigma}) &= A_i B_j (\delta_{ij} + i \epsilon_{ijk} \sigma_k) \\
 &= \vec{A} \cdot \vec{B} + i (\vec{A} \times \vec{B}) \cdot \vec{\sigma}
 \end{aligned}$$

(2) Suppose

$$(\vec{A} \cdot \vec{\sigma})(\vec{B} \cdot \vec{\sigma}) = m_0 + \sum_{i=1}^3 m_i \sigma_i$$

Then

$$\begin{aligned}
 m_0 &= \frac{1}{2} \text{Tr} [(\vec{A} \cdot \vec{\sigma})(\vec{B} \cdot \vec{\sigma})] \\
 &= \frac{1}{2} A_i B_j \text{Tr}(\sigma_i \sigma_j) \\
 &= \frac{1}{2} A_i B_j 2 \delta_{ij} = \vec{A} \cdot \vec{B}
 \end{aligned}$$

$$m_1 = \frac{1}{2} A_i B_j \text{Tr}(\sigma_i \sigma_j \sigma_1)$$

The trace is nonzero and equal to 2i only if $i=2, j=3$ or $i=3, j=2$

$$\begin{aligned}
 \Rightarrow m_1 &= i(A_2 B_3 - A_3 B_2) \\
 &= i(\vec{A} \times \vec{B})_1
 \end{aligned}$$

Similarly for m_2

Problem 5 (14.3.5)

Suppose a 2×2 matrix M is expanded as follows

$$M = \sum_{\mu=0}^3 m_{\mu} \sigma_{\mu} \quad \sigma_0 \equiv \mathbb{I}_{2 \times 2}$$

Then $\text{Tr } M = 2m_0$

since $\text{Tr } \sigma_i = 0 \quad i=1, 2, 3 \dots$

$$\text{Tr}(M \sigma_i) = m_0 \text{Tr } \sigma_i + \sum_j m_j \text{Tr}(\sigma_j \sigma_i)$$

Now use

$$\text{Tr } \sigma_i \sigma_j = 2 \delta_{ij}$$

$\Rightarrow$ and $\text{Tr } \sigma_i = 0$

$$\text{Tr}(M \sigma_i) = 2m_i$$

Thus

$$m_0 = \frac{1}{2} \text{Tr } M$$

$$m_i = \frac{1}{2} \text{Tr}(M \sigma_i)$$

Direct evaluation

$\Rightarrow$

$$m_0 = \frac{1}{2} (\alpha + \delta)$$

$$m_1 = \frac{1}{2} (\beta + \gamma)$$

$$m_2 = \frac{1}{2} i(\beta - \gamma)$$

$$m_3 = \frac{1}{2} (\alpha - \delta)$$

which may be solved for $\alpha, \beta, \gamma, \delta$

Problem 6 (14.3.7)

$$\begin{aligned}
 (1) \quad I + i\sigma_x &= \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \sigma_x \right) \\
 &= \sqrt{2} \left(\cos \frac{\pi}{4} + i \sigma_x \sin \frac{\pi}{4} \right) \\
 &= \sqrt{2} e^{i\sigma_x \pi/4}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thus} \quad (I + i\sigma_x)^{1/2} &= 2^{1/4} e^{i\sigma_x \pi/8} \\
 &= 2^{1/4} \left[\cos \frac{\pi}{8} + i \sigma_x \sin \frac{\pi}{8} \right]
 \end{aligned}$$

$$(2) \quad (2I + \sigma_x)^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

$$(2I + \sigma_x)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$= \frac{1}{3} (2I - \sigma_x)$$

$$(3) \quad \sigma_x^{-1} = \sigma_x \quad \text{since } \sigma_x^2 = I$$

7 ~~11~~ ~~12~~

Problem 5 (14.3.8)

Consider a general matrix

$$M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

Then, as in Problem 4

$$M = \sum_{\mu=0}^3 m_{\mu} \sigma_{\mu}$$

$$\begin{aligned} (1) \quad [M, \sigma_i] &= \sum_{\mu=0}^3 m_{\mu} [\sigma_{\mu}, \sigma_i] \\ &= \sum_{k=1}^3 m_k [\sigma_k, \sigma_i] \\ &= 2 \sum_{k=1}^3 m_k i \epsilon_{kij} \sigma_j \end{aligned}$$

Now, $\sum_{i=1}^3 m_i \sigma_i = 0 \Rightarrow m_i = 0 \quad \forall i$
Thus

$$\sum_k \epsilon_{jki} m_k = 0 \quad \forall (i,j)$$

$$[M, \sigma_1] = 0 \Rightarrow m_2 = m_3 = 0$$

$$[M, \sigma_2] = 0 \Rightarrow m_3 = m_1 = 0$$

Thus only $m_0 \neq 0$ possible
ie M must be a multiple
of the unit matrix

(2) For the anticommutator

$$\{M, \sigma_i\} = 2m_0 \sigma_i + \sum_j m_j \{\sigma_j, \sigma_i\}$$

However

$$\{\sigma_j, \sigma_i\} = 2\delta_{ji}$$

Thus

$$\{M, \sigma_i\} = 2(m_0 \sigma_i + m_i)$$

$$\text{Thus } \{M, \sigma_i\} = 0 \Rightarrow m_0, m_i = 0$$

e.g

$$\begin{aligned} 0 = \{M, \sigma_3\} &= 2(m_0 \sigma_3 + m_3) \\ &= 2 \begin{pmatrix} m_0 + m_3 & 0 \\ 0 & m_0 - m_3 \end{pmatrix} \end{aligned}$$

$$\Rightarrow m_0 = m_3 = 0$$

$$\{M, \sigma_1\} = 0 = 2(m_0 \sigma_1 + m_1)$$

$$\Rightarrow m_1 = 0$$

$$\{M, \sigma_2\} = 0 \Rightarrow m_2 = 0 \text{ as well } m_0 = 0$$

Thus such a M must vanish